

Speech-Driven Drafting of First Information Reports for the Tamil Nadu Police

Implementation progress and interim results: one LangGraph pipeline from a spoken Tamil complaint to a draft IF-1, with the officer as author of record.

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Guide

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16 September 2026

Review 3 rubric coverage

BCSE497J Review 3 — 7 parameters, 20 marks. Where each is answered in this deck.

#	Evaluation parameter	Marks	Slides	Evidence
1	Follow-up on Review 2 feedback; progress vs approved plan	3	3, 4	action table; 27-milestone status
2	Implementation and functional progress (~50% of scope)	3	5–11	end-to-end pipeline, live demo
3	Technical accuracy and best practices	3	14	368 tests, pinned invariants, gazette-cited tables
4	Interim testing, results and analysis	3	8, 9, 12	F1 / WER-CER / chrF / P-R tables and figures
5	Problem-solving and technical refinement	2	13	9 findings, each with the corrective action
6	Progress of the project report	3	15	Chapters 1–3 revised, Chapter 4 drafted
7	Presentation, individual responsibility, Q&A	3	16, 18	contribution table, demo, questions

Follow-up on Review 2 feedback

1 mark: action taken on panel comments. 2 marks: progress against the approved plan (next slide).

Review 2 comment / suggestion	Action taken	Where
«Panel comment 1 — copy from the Review 2 evaluation sheet»	«what was done»	«slide / section»
«Panel comment 2»	«what was done»	«slide / section»
«Panel comment 3»	«what was done»	«slide / section»

Decisions changed since Review 2 — and why

- Compute: the dev machine has an 8 GB GPU, not the 16 GB assumed — the plan's 14B synthetic-data model does not fit; the 7B / 14B-with-offload decision is measured, not assumed (finding 1).
- Statute classifier: InLegalBERT was to be the classifier; on ILSI a TF-IDF baseline beats it by 0.17 micro-F1 and an ensemble adds +0.0006 — TF-IDF is the classifier, the encoder's job becomes element verification (finding 8).
- Cognizability: the planned First Schedule table was a hand-coded stub; it is now parsed from the gazette PDF with page citations, and the stub turned out wrong on 4 of 48 entries (finding 9).
- A Tamil→English translation node was added — the classifier is English-only — and its first real failure (“dowry” → “relief”) produced the cue-scan safety net (finding 5c).

Progress against the approved plan (PLAN.md, 27 milestones)

10 working · 11 partial / v0 · 6 not started. Everything green or amber runs inside one end-to-end graph today.

PLAN.md milestones at Review 3: 10 working, 11 partial, 6 not started (of 27)

P0.1 Repo, env, GPU baseline	working
P0.2 IF-1 JSON Schema	working
P0.3 Legal KB: BNS text, First Schedule, IPC→BNS map	working
P0.4 Ethics / DPDP memo	partial / v0
P1.1 Audio front-end (VAD)	partial / v0
P1.2 ASR baseline + guards	working
P1.3 Tamil ITN	working
P1.4 Guided-interview TTS (optional)	not started
P2.1 Extraction A: rules (+NER later)	partial / v0
P2.2 Extraction B: LLM structured	not started
P2.3 Bilingual fields	partial / v0
P3.1 IPC→BNS mapping layer	working
P3.2 Multi-label statute classifier	working
P3.3 Element-wise verifier (rule v0; LLM later)	partial / v0
P3.4 Cognizability gate (gazette Schedule)	working
P4.1 Deterministic IF-1 template engine	working
P4.2 Grounded narrative (deterministic v0)	partial / v0
P4.3 Faithfulness gate	partial / v0
P4.4 Print-faithful PDF	not started
P5.1 Review workspace (static page v0)	partial / v0
P5.2 Statute panel	partial / v0
P5.3 Audit & versioning	not started
P6.1 End-to-end LangGraph orchestration	working
P6.2 Fine-tuning pass	not started
P6.3 Human gold set	not started
P6.4 Full evaluation study (harness exists)	partial / v0
P6.5 Write-up & demo	partial / v0

Semester-1 target (PLAN §2):

– “a thin end-to-end vertical slice on synthetic data”

Status:

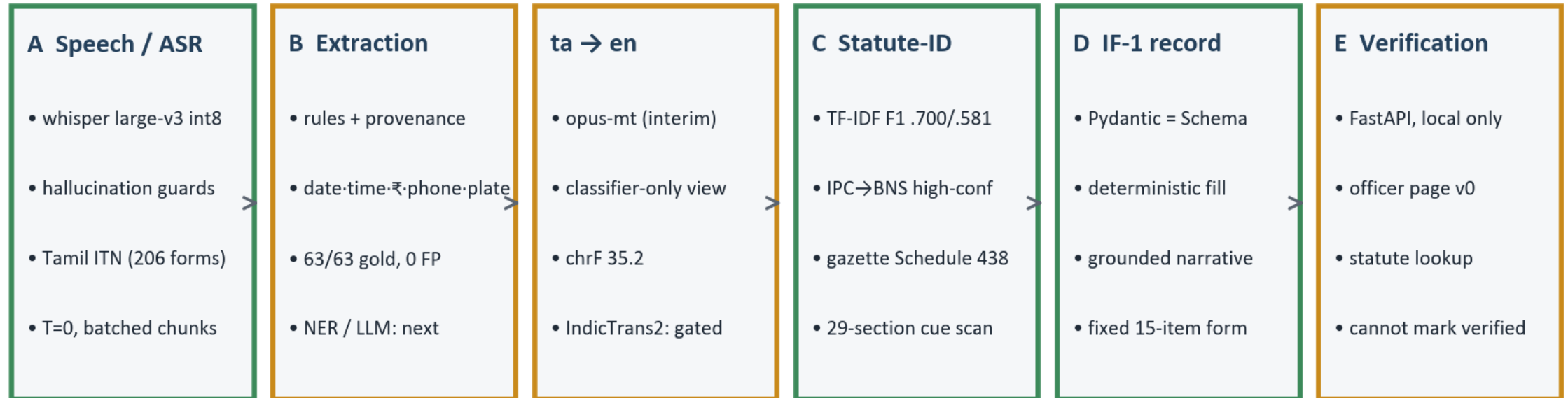
– the slice is not thin any more — ASR → extraction → translation → statute-ID → IPC→BNS → cognizability → elements → IF-1 → form → officer page run as one LangGraph graph

Deliberately not started:

– LLM-based modules (P2.2, P3.3-LLM, P4.2-LLM, P6.2) wait on the 8 GB model decision; PDF form and audit log are semester-2 items

What runs end to end today

One graph: (audio→asr) → extract → translate → statute_id → ipc_bns → cognizability_check → instantiate. ~10,400 lines; text or audio in, IF-1 draft out.



green = working end to end · amber = working v0, upgrade planned

– Decision support by construction: no endpoint, CLI or graph node can set a record to “verified”; the form prints “DRAFT — the officer is the author of record”; every machine-filled slot carries provenance (character offsets into the transcript).

– Strict local-only: models from a local HF cache; the API binds 127.0.0.1; audio uploads are deleted after transcription.

Stage A — Speech and ASR: baseline, guards, normalisation

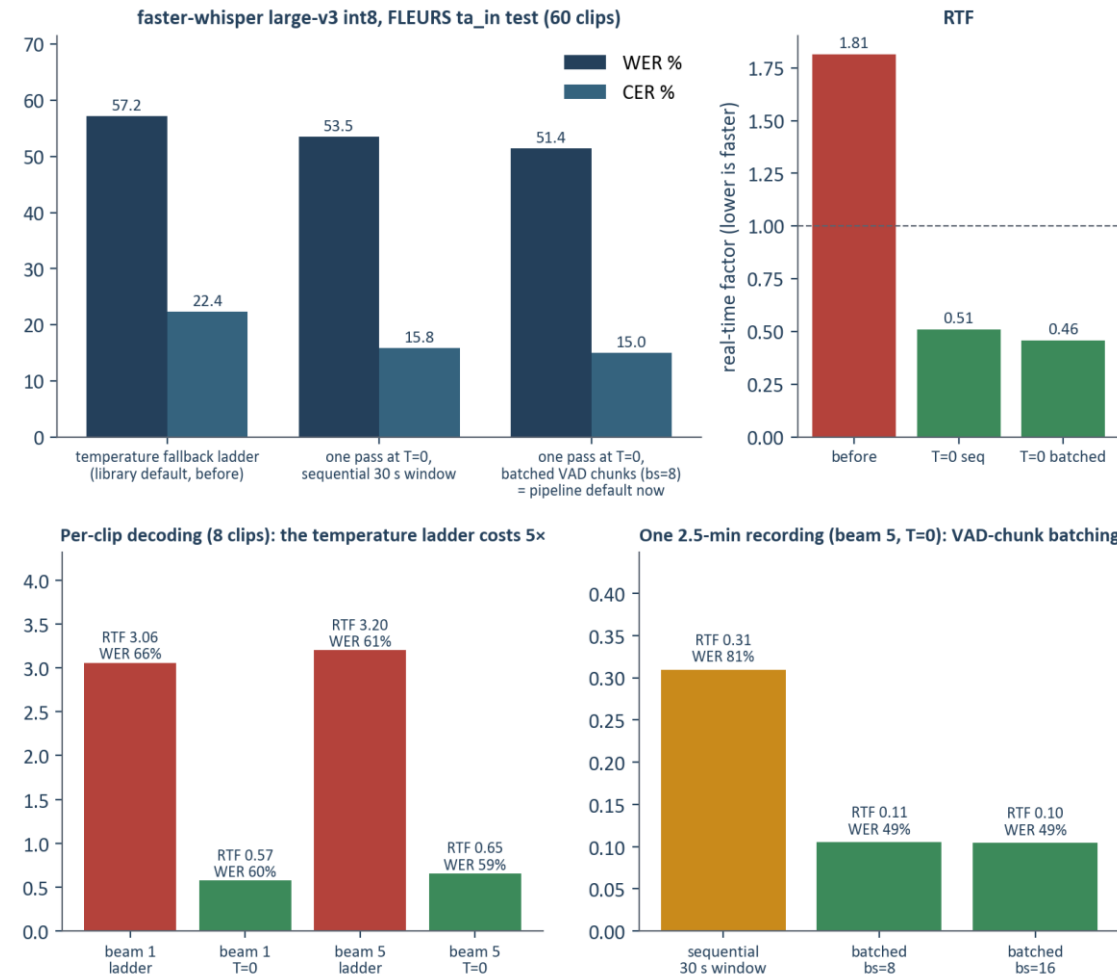
faster-whisper large-v3 (CTranslate2) on the 8 GB GPU; FLEURS ta_in as the zero-shot yardstick.

Built

- Zero-shot baseline on 60 clips: WER 57.2% → 51.4%, CER 22.4% → 15.0% after this week's decoding fix; Tamil is agglutinative, so CER is reported alongside WER (finding 6c)
- Hallucination guards: duplicated-trigram fraction (replay loops) and foreign-script count (language drift); thresholds calibrated on the real baseline — clean clips score exactly 0%; a flagged transcript blocks auto-resolution (finding 6b)
- Tamil inverse text normalisation: 206-form number lexicon with sandhi, spoken dates/times/amounts → digits, offset map back to the audio transcript (69 tests)

Fixed this week (finding 6f)

- The library's temperature-fallback ladder re-decoded low-confidence Tamil windows up to six times. One pass at T=0 + batched VAD chunks: on 60 clips RTF 1.81 → 0.46 and WER/CER improved, not traded
- On a complaint-length recording: RTF 0.31 → 0.10, WER 81% → 49% (chunks follow pauses, not fixed 30 s cuts)
- Honest caveat: 2 of 60 clips still trip the repetition guard at T=0 — the ladder was the speed cost, not the cause of the replay loop; the guard stays



Stage B — Extraction floor, and the Tamil→English view

Rules with provenance now; NER / LLM extraction is the semester-2 upgrade. The classifier is English-only, so Tamil input is translated for it — and only for it.

Rule extractor (Stage B floor)

- Slots: dates (digit, spoken, year-less), times, ₹ amounts (digit and spoken, Indian grouping), phone numbers, vehicle plates
- Fills a slot only when unambiguous; several candidates → blank + officer note
- Gold smoke set: 28 adversarial complaints, 63/63 slots, 0 false positives
- Every value carries character offsets — the form can point back at the words
- Limit: names, places and accused descriptions are not extracted yet — that is what the synthetic-data + NER phase is for

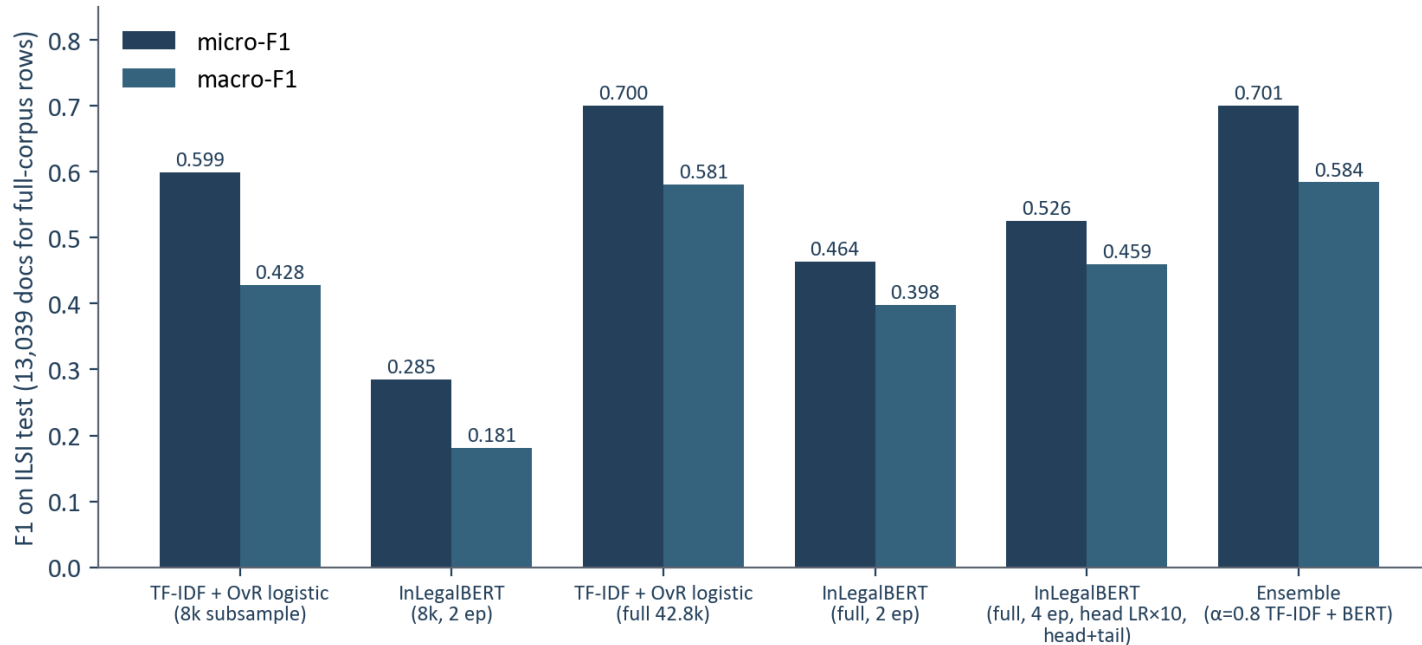
Translation node (finding 5b / 5c)

- Interim model opus-mt-dra-en: chrF 35.2, BLEU 10.5 on FLoRes ta→en; IndicTrans2 (~+20 chrF) is gated behind a licence click
- Output feeds the classifier only; extraction, elements and the narrative keep the original Tamil
- First real failure: “வரதட்சணை” (dowry) → “relief”; the classifier saw a threat, and a 498A complaint routed to CSR
- Fix: a bilingual element cue-scan on the ORIGINAL text; a cognizable section whose ingredients are all present but unpredicted forces officer review, naming the section
- Pinned by tests that reproduce the exact failure

Stage C — Statute identification: the encoder question is settled

ILSI: 42,835 training / 13,039 test court documents, 100 IPC sections, multi-label. Threshold tuned on dev.

Statute identification: a sparse baseline beats the legal encoder — and the ensemble adds nothing



Result

- TF-IDF + one-vs-rest logistic: micro-F1 0.700, macro-F1 0.581
- InLegalBERT with every lever (4 epochs, head LRx10, head+tail truncation, pos-weight): 0.526 / 0.459
- Ensemble gain over the best single model: +0.0006

Ruled out as causes

- truncation to 512 wordpieces (costs TF-IDF only 8.6% relative); head collapse (fixed with pos-weight, finding 7); too little data (full corpus used)

Decision

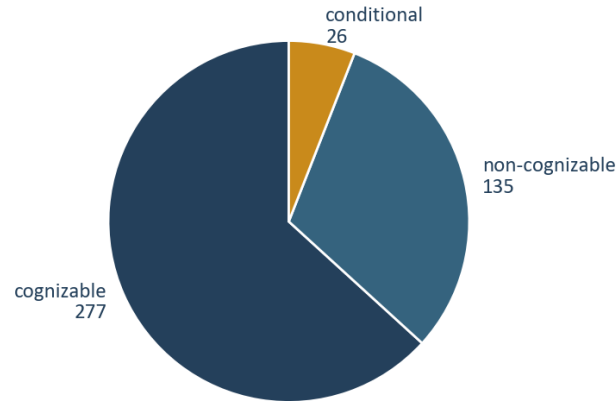
- TF-IDF is the classifier; the encoder's job is element verification. Keep the sparse floor in every comparison.

Stage C — The legal knowledge base is now the gazette, not memory

IPC→BNS map with a legal-safety invariant; BNSS First Schedule and all 358 BNS section texts parsed from the official gazette PDFs with page citations.

BNSS First Schedule Pt I parsed: 438 section keys

Where the hand-coded stub was wrong (4 refusal-direction errors, 2 hidden conditions)



BNS section	hand-coded stub said	gazette Schedule says
126 wrongful restraint	non-cog.	cognizable
223 disobeying public serv	non-cog.	cognizable
296 obscene acts	non-cog.	cognizable
329 criminal trespass	non-cog.	cognizable
85 cruelty (498A)	cognizable	conditional*
303(2) theft	cognizable	conditional†

* cognizable only when reported by the aggrieved woman / relative † non-cognizable below ₹5,000

IPC→BNS map (100 ILSI sections)

– 69 auto-applicable · 29 flagged for review · 7 map outside the BNS. Invariant, pinned by tests: only confidence = high AND needs_review = N AND a real BNS target is applied without a human.

BNSS First Schedule, Part I

– 438 section keys: 277 cognizable, 135 non-cognizable, 26 conditional (abetment follows the offence; s.85 depends on who reports; s.303(2) theft splits at ₹5,000 — resolved from the extracted value)

BNS 2023 text

– 358 sections with headings, chapters and illustration-free operative text; shown beside every suggestion, IPC text kept as lineage
– Status: parsed and spot-checked, not legally signed off

Stage C — Element-wise justification (v0) and the cue-scan safety net

A section is not one thing. The officer sees which ingredients the facts show, quoted from the transcript — and which are still to be established.

Output on the form (IPC 304B → BNS 80, dowry-death sample)

- [?] death of a woman — not stated in the narrative; establish: is the deceased a woman?
- [x] by burns, bodily injury, or otherwise than in normal circumstances · 'murder'
- [x] within seven years of marriage · 'married five years ago'
- [x] cruelty or harassment for dowry, soon before death · 'cruelty'; 'demand'; 'dowry'

- Two rules keep it honest: never “no” (absence of a word is not evidence of absence — it yields “unclear”, the officer's cue to ask), and every “yes” carries the matched words and offsets
- 29 sections covered with English + Tamil cues (theft, robbery, hurt, kidnapping, cheating incl. OTP fraud, cruelty, dowry death, trespass ...); a 22-complaint battery pins what must and must not fire

Why it doubles as a safety net

- The classifier reads an English view of the complaint; the cues read the original Tamil
- A covered section whose every ingredient is present but which the classifier did not predict is raised as a question — never auto-applied
- If that section is cognizable or conditional and the route would otherwise be CSR, the case goes to the officer with the section named
- Next (PLAN P3.3): replace keyword cues with a local LLM reading each element's BNS text against the complaint — same schema, same tests

Stages D and E — IF-1 record, grounded narrative, form, officer page

Deterministic slot-filling into the fixed CCTNS IF-1; the only free text is the narrative box, and every sentence in it must trace to an extracted fact.

Stage D

- Pydantic record = the canonical IF-1 JSON Schema (parity tests fail on any drift); status draft → csr_advisory → verified, and nothing in src/ can write “verified”
- Decision → record bridge: auto-applied sections become BNS candidates with IPC lineage; held-back mappings are kept, flagged ambiguous, score-capped
- Narrative composer: one sentence per grounded fact, Tamil + English, faithfulness gate = every sentence supported (structural v0; NLI/LLM judge later)
- Renderer: the 15-item form, machine-filled slots annotated ⟨status · confidence · source⟩, items 13–15 always blank (they are the officer's)

Stage E

- FastAPI on 127.0.0.1: /v1/complaint/{text,audio}, /v1/statute/{section}, /v1/health; a static officer page: route, sections with BNS text and element checks, “also consider”, flags, annotated form; “Mark verified” disabled by design

Live demo (5 min)

- 1 Typed theft complaint with a value → FIR route, BNS 303(2) with the ₹5,000 proviso shown
- 2 Same complaint without the value → officer review: “cognizability depends on property value”
- 3 Tamil dowry-cruelty complaint → translation loses “dowry”; the cue scan still raises IPC 498A
- 4 Look up BNS 85 → gazette text + the Schedule's condition
- 5 Spoken Tamil clip → transcript with ITN, guards, then the same path
- Fallback if the GPU is busy: `python -m fir draft --json` on the CLI, same output

Interim results at a glance

All numbers are read from artifacts/reports/.json by the build script; RESULTS.md is regenerated by `make results`.*

Component	Data	Metric	Result	Read
Statute-ID (TF-IDF)	ILSI test, 13,039 docs	micro / macro-F1	0.700 / 0.581	the classifier of record
Statute-ID (InLegalBERT, best)	same	micro / macro-F1	0.526 / 0.459	loses to the sparse floor
ASR before (fallback ladder)	FLEURS ta_in, 60 clips	WER / CER / RTF	57.2% / 22.4% / 1.81	2 clips flagged by the guard
ASR now (T=0, batched)	same	WER / CER / RTF	51.4% / 15.0% / 0.46	2 flagged: replay loop persists at T=0
ASR (T=0, sequential)	same	WER / CER / RTF	53.5% / 15.8% / 0.51	1 flagged
ASR, complaint-length recording	8 clips joined, 148 s	RTF / WER	0.10 / 49% (batched) vs 0.31 / 81% (sequential)	batching is also more accurate
Translation ta→en (opus-mt)	FLoRes devtest, 336 pairs	chrF / BLEU	35.2 / 10.5	upper bound on Tamil statute-ID
Extraction (rules)	28 gold complaints	slot P / R	1.00 / 1.00 (63 slots, 0 FP)	smoke set, not a benchmark
IPC→BNS map	100 ILSI sections	auto / review / non-BNS	69 / 29 / 7	third-party source; gazette check pending
BNSS First Schedule	438 keys from gazette	cog / non-cog / conditional	277 / 135 / 26	stub was wrong on 4/48
Test suite	17 files	tests passing	368	no model weights needed, ~20 s

Technical challenges and corrective action

Each finding is recorded in PROGRESS.md with the measurement that established it and the test that keeps it fixed.

Challenge (finding)	Evidence	Corrective action
GPU is 8 GB, not 16 GB (1)	torch reports 8.5 GB; 14B-Q4 ≈ 9 GB	models.yaml carries a fits flag per model; 7B vs 14B-with-offload to be benchmarked on mains power
WSL VM rebooted under full-corpus jobs (6e)	6 deaths; predictor = Python heap > ~2 GB	chunked vectoriser / tokenizer / predict; detached launches; logs on /mnt/c
InLegalBERT head collapsed to 6/100 labels (7)	loss flat at 0.142, macro-F1 0.018	pos-weighted BCE, higher head LR; then measured honestly against TF-IDF
Whisper fabricated text on 2/60 clips (6b)	replay loop 226% WER; Cyrillic drift	transcript guards calibrated on the baseline; flagged → not auto-resolvable
ASR 3× slower than real time (6f)	profile: fallback ladder 3.06 vs 0.57 RTF	one pass at T=0; batched VAD chunks; 60-clip WER 57→51%, CER 22→15%, RTF 1.81→0.46
Translation lost “dowry” → CSR route (5c)	reproduced on the real server	bilingual cue scan on the original; forces officer review
Cognizability stub wrong on 4/48 (9)	gazette parse vs stub: 126, 223, 296, 329	parse the First Schedule with page citations; conditional class; value-resolved theft split
Classifier abstains on short complaints (5)	top-5 for a theft sample has no 379	not a threshold problem — in-domain synthetic data; cue scan carries short inputs

Technical accuracy and engineering practice

What a reviewer can check without trusting us.

Testing

- 368 tests in 17 files, ~20 s, no weights: golden end-to-end fixtures, schema parity, ITN, cue battery, Schedule integrity, serving, CLI
- Legal invariants are tests: the IPC→BNS auto-apply rule, “unknown never defaults”, “nothing can set verified”, threat ≠ attempt/murder
- ruff clean; LF-only repo; reproducible builds: `make schedule`, `make bns-text`, `make results`

Evaluation practice

- Sparse baseline kept in every comparison; thresholds tuned on dev, reported on test; CER beside WER for Tamil; per-clip distributions, not just corpus means
- Negative results recorded: encoder < TF-IDF; ensemble +0.0006; condition_on_previous_text did not fix replay; int8 vs fp16 WER gap was fallback noise

Provenance and safety

- Every extracted value: character offsets; every element “yes”: quoted words; every Schedule row: gazette page; every model: local cache path
- Decision support only — route and sections are proposals; the officer is the author of record; CSR is advisory
- Local-only inference; no hosted API anywhere in the code; PII-minimal (audio deleted after transcription)

Best-practice choices

- CTranslate2 int8 for VRAM headroom; LangGraph for an inspectable state machine; Pydantic ↔ JSON Schema parity; config-driven (pipeline.yaml, models.yaml, datasets.yaml)
- Honest limits stated on the form: “stub”/“parsed, not signed off”/“interim model” wherever true

Project report progress

Review3_Report.docx: Chapters 1–3 carried forward with a revisions section; Chapter 4 drafted from the same JSON reports as this deck.

Chapter	Status	Contents at Review 3
Preliminary pages	updated	abstract with a Review-3 status paragraph; rubric map for Review 3; TOC
1 Introduction	unchanged	background, problem, objectives, scope, expected outcomes
2 Literature / patent review	unchanged	16 sources; comparative analysis; gap
3 Methodology / design	revised	§3.8 Revisions since Review 2: compute budget, classifier choice, translation node, gazette-parsed KB, team allocation
4 Implementation and results	new (initial)	environment; implementation by stage; testing & validation; interim results (6 tables, 5 figures); analysis; limitations; individual contribution
5 Conclusion and future work	outline	to be written for Review 4
References	extended	gazette notifications, faster-whisper / CTranslate2, opus-mt, sacreBLEU added

AI-tool acknowledgement (guideline §2): an AI coding assistant (Claude Code) was used for implementation support under the team's direction; all design decisions, legal tables and results were reviewed and are reproducible from the repository. «confirm wording with the guide»

Individual contribution

«Confirm before the review — the panel grades each member's demonstrated contribution (guideline §2).»

Member	Ownership (as planned at Review 2)	Delivered at Review 3	Demonstrates
Madhav K (23BAI1088)	Stage C/D: statute identification, IPC→BNS, cognizability, IF-1 instantiation; integration	«...»	«slides / demo steps»
Nischay Kuchibotla (23BAI1245)	Stage A: ASR benchmark, guards, Tamil ITN, code-switch fine-tune	«...»	«...»
Rohit A. S. (23BAI1416)	Stage B: extraction, synthetic data engine, NER seeding	«...»	«...»

The repository records who did what: PROGRESS.md is the dated engineering log; every module has an owner in IMPLEMENTATION.md. Present the split as it actually happened — the rubric rewards demonstrated contribution, not the plan.

Plan to Review 4 (12–16 October 2026)

Complete implementation, final results, report verification — and proof of a publication submission (Review 4 parameter 9, 2 marks).

Weeks 1–2

- Decide the local LLM on mains power (7B-Q4 vs 14B-Q4 offload) → synthetic Tamil complaint → IF-1 generator (PLAN §5.2)
- IndicTrans2 (licence click) or NLLB-600M replaces opus-mt; re-measure Tamil statute-ID end to end
- IndicWhisper / IndicConformer benchmark against the corrected Whisper numbers

Weeks 3–4

- LLM element verifier reading BNS text (P3.3) behind the same ElementCheck schema; NER (MuRIL) for names / places / accused
- Legal read-through of the Schedule CSV and the 29 flagged mapping rows with the guide
- Self-recorded Tamil complaint gold set (P6.3), consent script from the DPDP memo

Publication plan «confirm venue with the guide»

- Paper 1 (ready to write from Chapter 4): “A sparse baseline beats a domain-pretrained encoder on Indian statute identification — and what the encoder is for instead” (ILSI, full-corpus results, element verification)
- Paper 2 (after the gold set): “Officer-in-the-loop FIR drafting from spoken Tamil complaints: provenance, cognizability and the cost of translation”
- Candidate venues: ICON 2026 (NLP in India), FIRE 2026 (legal track), or an IEEE/Springer conference with a Dec–Jan deadline — submission proof needed by 12 Oct
- Artefacts already reusable: RESULTS.md tables, r3_*.png figures, reproducible harness

Summary

Half the approved scope runs as working modules, in one graph, with the officer as author of record.

- An end-to-end pipeline from spoken Tamil to a draft IF-1 exists and is tested; nothing in it can file anything
- The research question was measured, not assumed: on ILSI a sparse baseline beats InLegalBERT (0.700 vs 0.526 micro-F1); the encoder's role is element verification
- The legal tables the routing depends on now come from the gazette with page citations — and doing so caught four refusal-direction errors in the hand-coded version
- ASR runs 4× faster with better WER and CER after one measured change; two real safety failures (translation loss, fabricated transcripts) each produced a guard that is pinned by a test
- Next: the local LLM decision unblocks synthetic data, the element verifier and the fine-tuning pass; publication draft from Chapter 4

Questions — and the live demo.